DR-2 lattice closure mapped onto STEP-5B G1-triple-prime-AE – the

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1. Purpose and scope

R-026 closes the (unweighted) additive-energy DR-2 for the crystallographic lattice class at T7. STEP-5B is CLOSED-CONDITIONAL on H-ADM-COH, whose role is to make the G1”’-AE corner – the weighted sphere additive-energy bound – hold by restricting to angularly SEPARATED competitor patterns. This note maps the lattice closure onto G1”’-AE through an amplitude bridge and analyses what remains for an H-ADM-COH discharge. It introduces no new physics; it is a reduction/integration note.

2. Three identifications

(i) G1”’-AE is a weighted additive energy. The STEP-5B corner requires $\sum_t w_t^2 \le K(\lambda' I)^2$ with $w_t = \sum_{a+b=t} c_a c_b$ the amplitude-weighted representation function of the pattern points a, b on the shell and $I = \|c\|_2^2$. This is the weighted additive energy of the pattern.

(ii) The “carrier” is the sum-level circle. A pair (a, b) with $a + b = t$ is diametral on its carrier circle of centre $t/2$; that carrier is exactly the sum-level circle $C_t = \{x : |x|^2 = R,\ x \cdot t = \frac{1}{2}|t|^2\}$ of R-025. The carrier occupancy is $\#(Q \cap C_t)$, whose maximum is $T'(Q)$, and the extraction-route carrier-richness $\chi(P)$ is the dual incidence quantity (rich carriers through a point).

(iii) H-ADM-COH is angular separation. H-ADM-COH admits competitor patterns whose pairwise angular separations are $\ge \theta_{\min}(\hat{r})$; finer-scale mass is assigned to the Gaussian sea. This is the SEPARATED-pattern restriction – the regime in which the additive-energy bound is classically easy. Removing H-ADM-COH is the unrestricted (non-separated) problem.

3. The amplitude bridge (WEIGHTED Lemma A)

Lemma (weighted). For finite $Q \subset S^2$ and any amplitudes $c = (c_a)_{a \in Q}$,

$$\boxed{\sum_t w_t^2 \ \le\ \big(1 + T'(Q)\big)\, \|c\|_2^4}, \qquad w_t = \sum_{a+b=t} c_a c_b.$$

Proof. Split $\sum_t w_t^2 = w_0^2 + \sum_{t \ne 0} w_t^2$. For $t = 0$, $w_0 = \sum_{a : -a \in Q} c_a c_{-a}$, so by Cauchy–Schwarz $|w_0| \le \big(\sum_a c_a^2\big)^{1/2}\big(\sum_a c_{-a}^2\big)^{1/2} = \|c\|_2^2$, i.e. $w_0^2 \le \|c\|_2^4$. For $t \ne 0$, Cauchy–Schwarz over the $r(t)$ summands gives $w_t^2 \le r(t) \sum_{a+b=t} c_a^2 c_b^2$, and $r(t) \le \#(Q \cap C_t) \le T'(Q)$, so

$$\sum_{t \ne 0} w_t^2 \le T'(Q) \sum_{t \ne 0} \sum_{a+b=t} c_a^2 c_b^2 \le T'(Q) \sum_{a,b} c_a^2 c_b^2 = T'(Q)\, \|c\|_2^4.$$

Adding the two parts gives the bound. □

This is the amplitude-weighted analogue of Lemma A (R-025): the SAME sum-circle richness T' controls the weighted energy, for arbitrary amplitudes – so the amplitude degree of freedom of G1"'-AE does not require new geometry.

4. The integration bound

Theorem. Let $Q \subseteq \Lambda \cap \{|x|^2 = R\}$ be any subset of a crystallographic momentum shell, and c any amplitudes. Then

$$\sum_t w_t^2 \;\le\; \big(1 + 6 \max_{t\neq 0} d(4R - |t|^2)\big)\|c\|_2^4 \;\le\; \big(1 + C_\epsilon R^\epsilon\big)\|c\|_2^4.$$

Equivalently, $\sum_t w_t^2 \le K\,\|c\|_2^4$ with $K = 1 + C_\epsilon R^\epsilon$ – the G1"'-AE bound with a SUBPOLYNOMIAL constant, holding for ALL subsets of the shell (no angular separation).

Proof. WEIGHTED Lemma A bounds $\sum_t w_t^2$ by $(1 + T'(Q))\|c\|_2^4$; R-026 ([DIV-CIRC]) bounds $T'(Q) \le 6 \max_{t\neq 0} d(4R - |t|^2) \ll_\epsilon R^\epsilon$. □

5. What this means for H-ADM-COH (operator decision)

The bound of §4 is exactly the G1"'-AE requirement, established for the UNRESTRICTED (non-separated) lattice class. Since H-ADM-COH was the angular-separation restriction that secured G1"'-AE, the lattice arithmetic REPLACES separation: for the physical (crystallographic-shell) competitor class, the weighted additive-energy bound holds without H-ADM-COH. The discharge is nonetheless NOT enacted here; three residuals, all operator-level, remain:

(a) *Subpolynomial vs constant K.* §4 gives $K = 1 + C_\epsilon R^\epsilon$, not a fixed constant. With $R \asymp N^2$, $K \sim N^{2\epsilon}$ (measured $K = 1 + T'$ grows like $R^{0.126}$). This matches the pre-registered sharp-$O(N^2)$ conjecture (exponents 2.04–2.08) and is far inside the $\times 55.6/\times 8.8/\times 2.1$ margins for the operating n, but whether the gate accepts a subpolynomial K in place of the constant $K = 14$ form is the gate-keeper's call.

(b) *Modeling identification.* The discharge requires that admissible competitor patterns are exactly subsets of the crystallographic momentum shell (physical for a lattice field theory); this is an operator modeling confirmation, not a mathematical step.

(c) *Corner coverage.* The G1"'-AE corner (non-transversal multi-circle high-n) is a lattice-shell subset, hence covered by §4; the operator confirms no corner escapes the lattice identification.

If (a)–(c) are accepted, H-ADM-COH may be discharged for the lattice class and B1's hypothesis set reduced accordingly. This note makes NO such flip: DR2-SHARE stays OPEN, B5 T5, B1 T6, H-ADM-COH retained.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) *"WEIGHTED Lemma A might fail for signed/complex amplitudes."* Checked: the Cauchy–Schwarz proof is amplitude-agnostic; verified on uniform, Gaussian, peaked, and signed profiles (worst ratio $0.277 \le 1$, dr2_weighted_energy.py 3/3).

(β) *"This silently flips H-ADM-COH."* It does NOT. §5 lists three operator-level residuals and makes no gate/tier/hypothesis change. The note's theorems (WEIGHTED Lemma A, the §4 bound) are T7; the discharge is deferred.

(γ) *"K is subpolynomial, so this is weaker than the constant-K G1"'-AE."* VALID and recorded as residual (a). A subpolynomial K is consistent with the sharp-$O(N^2)$ conjecture and the measured exponents; whether it suffices for the margin closure is the operator's call, not a mathematical defect of the bound.

(δ) *"The $\chi(P)$ route is bypassed – is that legitimate?"* YES: the extraction route's $\chi(P)$ obstruction (DR2-SHARE) was internal to one proof strategy. The lattice bound proves the additive energy DIRECTLY (Lemma A + divisor bound), so the weighted energy is controlled without ever iterating the extraction; $\chi(P)$ is not needed. DR2-SHARE (the extraction obstruction) is thereby rendered moot FOR THE LATTICE CLASS, though it stays formally OPEN.

Quantitative sanity check (mandatory). WEIGHTED Lemma A verified exactly: $\sum_t w_t^2 \leq (1+T')\|c\|_2^4$ on shells $R = 101, 314, 561$, four amplitude profiles, worst ratio 0.277. Antipodal term $w_0^2 \leq \|c\|_2^4$ checked. Subpolynomial K: $1+T'$ grows like $R^{0.126} < R^{0.30}$. Direction: upper bound, $\|c\|_2^2 = I > 0$.

7. Result footer

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Result ID:                R-027 / DR-2 -> STEP-5B G1'''-AE integration
Precise statement:        WEIGHTED Lemma A: for finite Q in S^2 and any
                          amplitudes c, sum_t w_t^2 <= (1+T'(Q))||c||_2^4,
                          w_t = sum_{a+b=t} c_a c_b (Cauchy-Schwarz; t=0 split).
                          With R-026 (lattice T' <<_eps R^eps): for any subset Q
                          of a crystallographic shell, sum_t w_t^2 <=
                          (1+C_eps R^eps)||c||_2^4 -- the G1'''-AE bound with
                          K ~ R^eps, NO angular separation.
Scope:                    STEP-5B G1'''-AE corner; the DR2-SHARE integration;
                          the physical (lattice) competitor class.
Dependencies:             WEIGHTED Lemma A (T7, this note) + R-026 (T7).
                          Both unconditional modulo textbook number theory.
Evidence grade:           T7 (weighted Lemma A + integration bound). The
                          H-ADM-COH discharge is NOT enacted (operator decision,
                          residuals (a)-(c)).
Reproduction command:     python codes/vacuum/dr2_weighted_energy.py
Expected output:          3/3 PASS; sum_t w_t^2 <= (1+T')||c||_2^4 (worst ratio
                          0.277); w_0^2 <= ||c||_2^4; K=1+T' ~ R^0.126.
Falsification gate:       Amplitudes c with sum_t w_t^2 > (1+T'(Q))||c||_2^4
                          (refutes WEIGHTED Lemma A); not observed.
Tier before / after:      NEW T7 integration result. NO flip: H-ADM-COH retained,
                          DR2-SHARE OPEN, B5 T5, B1 T6.
No-overclaim statement:   Establishes the weighted G1'''-AE bound for the lattice
                          class with subpolynomial K, decoupling-free. Does NOT
                          discharge H-ADM-COH or change any tier; residuals
                          (a) subpolynomial-K acceptance, (b) modeling
                          identification, (c) corner coverage are operator
                          decisions. Arbitrary-Q (non-lattice) DR-2 stays OPEN.
Next required action:     Operator: accept/deny residuals (a)-(c); if accepted,
                          discharge H-ADM-COH for the lattice class and re-tier B1.
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